

Characteristics of Forces — Paper A

MEDIUM • 40 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer **every** question in the space provided. Show your working — method marks are awarded even when the final answer is wrong.
- A calculator is allowed, but every number on this paper is designed to work without one.
- Every answer must carry a **unit**, and every resultant force must carry a **direction**. An answer with no unit does not score the final mark.
- The mark for each question is shown in square brackets on the right.
- Reminder: **weight = mass × gravitational field strength** ($W = mg$), and on Earth **$g = 10 \text{ N/kg}$** .

SECTION 1 – NAMING FORCES

1. State what is meant by a **force**, and give the unit in which force is measured. [2]

2. Name the force being described in each case.

- (a) The force that opposes a car as it moves through the air. [1]

- (b) The force that pulls a hanging mass downwards. [1]

- (c) The force in a rope that is being pulled tight. [1]

3. State whether each force below is a **contact** force or a **non-contact** force.

- (a) Friction [1]

- (b) Magnetic attraction [1]

(c) Upthrust

[1]

4. State the **three** things a force can change about an object.

[3]

SECTION 2 – MASS AND WEIGHT

5. Calculate the weight of each object on Earth. Take $g = 10 \text{ N/kg}$.

(a) A bag of mass 5 kg.

[1]

(b) A child of mass 12 kg.

[1]

(c) A phone of mass 0.5 kg.

[2]

6. Calculate the mass of each object. Take $g = 10 \text{ N/kg}$.

(a) A crate that weighs 300 N.

[1]

(b) A tin that weighs 45 N.

[1]

(c) An apple that weighs 7 N.

[1]

7. Explain the difference between **mass** and **weight**. Your answer must include the unit of each. [3]

SECTION 3 – ADDING FORCES UP

8. For each pair of horizontal forces, calculate the resultant force. Give a direction as well as a size.

- (a) 30 N to the right and 20 N to the left. [1]

- (b) 45 N to the right and 45 N to the left. [1]

- (c) 12 N to the left and 30 N to the left. [2]

9. A book rests on a table. Its weight is 60 N downwards and the table pushes up on it with 60 N.

- (a) State the resultant vertical force on the book. [1]

- (b) Describe the motion of the book. [1]

- (c) State the word used to describe forces that add up to zero. [1]

- 10.** A car engine pushes the car forwards with 2400 N. Friction and air resistance together push backwards with 2400 N. Describe the motion of the car and explain your answer. [2]

End of paper. Check every answer has a unit or a form the question actually asked for.